



KENNY NGUYEN

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About Me

3rd year University of Toronto Information Science undergraduate. Innovative Full stack Engineer specializing in developing AI-driven applications and scalable architecture. Demonstrated expertise in Python, complemented by a strong focus on user experience and responsive design. Proven ability with hands-on projects to enhance productivity through effective communication and collaboration, consistently delivering high-quality products that align with client needs and deadlines both Front End and Back End.

Skills

- **Python Backend Development:** Specialized in developing scalable machine learning pipelines and backend solutions, processing and analyzing datasets exceeding 1M+ entries with precision.
- **Machine Learning & AI:** Expertise in building predictive models, data preprocessing, and deployment using Scikit-learn, TensorFlow, and Jupyter Notebook, delivering actionable insights.
- **Test Automation with Selenium:** Mastered in creating resilient test frameworks that improve efficiency, reduce QA cycles by 50%, and ensure robust software performance.
- **Modern Web Development:** Skilled in developing dynamic and responsive web applications using React.js, Node.js, and Next.js, ensuring seamless performance and high user engagement.
- **Database Design & Optimization:** Proficient in MySQL and PostgreSQL, implementing efficient database schemas and optimizing queries for large-scale applications.
- **API Development & Integration:** Adept at creating and integrating RESTful and GraphQL APIs, ensuring secure and scalable backend functionality.
- **Version Control & Collaboration:** Extensive experience with Git and GitHub, maintaining version integrity and fostering team synergy in agile development environments.
- **CI/CD pipeline:** Deep understanding of Agile and DevOps practices, streamlining development lifecycles to deliver high-quality software on time. Experienced in CI/CD tools such as Github Actions, Confluence, Trello.

Experience

HealthBridgeAI

<https://healthbridgeai.com/>

Develop, Test, and Optimize Backend Services

Architect and maintain scalable, high-performance RESTful APIs using FastAPI.
Implement JWT authentication with Keycloak, ensuring secure access control.

Fulltime AI Backend Engineer

01/2025 – 07/2025

AI & Machine Learning Model Integration

Collaborate with AI/ML teams to deploy LLM-powered NLP features into backend services.

Utilize LangChain and OpenAI APIs for AI-driven healthcare automation.

Implement backend support for AI-powered text generation, speech-to-text, and chatbot functionalities.

Scalability & Deployment with CI/CD

Develop and containerize backend services using Docker and deploy them efficiently across cloud-based infrastructure.

Automate CI/CD pipelines with GitHub Actions to ensure continuous deployment and integration.

Deploy microservices using Uvicorn with Gunicorn, optimizing for high-concurrency environments.

Secure API Gateway & Authentication

Design and implement OAuth2-based authentication via KeyCloak, ensuring secure access control mechanisms.

Integrate RBAC (Role-Based Access Control) to manage different permission levels in healthcare applications. Ensure compliance with industry security standards such as HIPAA & GDPR.

Netpalm

Software

Developer Intern

10/2024 – 02/2025

- **Implement, Test, and Debug Front-End Features:**

Utilize modern frameworks (using React and Umi.js) to build and refine user-facing components on the HealthBridgeAI.com platform. Employ testing frameworks (Selenium) to detect regressions early, track issues via integrated CI/CD pipelines, and quickly patch front-end bugs. Collaborate with QA and UX teams to maintain an intuitive, high-performing user experience.

- **Develop and Maintain Web Crawlers:**

Create and optimize robust web crawler solutions using tools like Python (Scrapy) to systematically extract, validate, and store high-value data. Ensure crawlers remain resilient in changing web landscapes by implementing error handling, rate limiting, and other best practices.

- **Perform In-Depth User Data Analysis:**

Cleanse, process, and analyze complex datasets via Python libraries (pandas, NumPy, scikit-learn). Provide actionable insights to project stakeholders by creating clear visualizations and summaries. When necessary, integrate results with cloud-based services (AWS) on Ubuntu Virtual Machine for scalable computation.

- **Contribute to Full-Stack and AI/ML Feature Development:**

Collaborate in Agile sprints to design and implement new software features for the HealthBridgeAI platform. Work with microservice architectures (Docker) and adopt DevOps best practices (Git-based workflows, CI/CD pipelines) to ensure smooth deployments. If required by the manager or team lead, explore and integrate AI/ML models using frameworks like TensorFlow, PyTorch, or scikit-learn—including data preprocessing, model training, and performance tuning.

Algorithmics - The International Coding School

Ho Chi Minh City, Vietnam

Programming

Instructor

06/2024 - 09/2024

- Designed and delivered engaging lesson plans aligned with curriculum objectives, incorporating interactive coding exercises to enhance student comprehension and retention.

- Assessed the performance of 20+ students through tailored tests and quizzes, providing actionable feedback that improved overall class performance by 20%. Provided individualized instruction to meet the needs of all students.

- Delivered personalized instruction to accommodate diverse learning styles, resulting in 90% student satisfaction based on post-course feedback surveys.

ERA GROUP

Ho Chi Minh City,
Vietnam

- Engineered website architecture and designed user-friendly interfaces using HTML, CSS, and JavaScript, enhancing user engagement by 25%. Created content management systems to manage websites efficiently.

Website Developer

11/2023 - 01/2024

- Built robust content management systems using WordPress and PHP, enabling efficient website updates, and reducing content deployment time by 40%.
- Integrated web applications with MySQL databases, REST APIs, and third-party services, ensuring seamless data exchange and system reliability across all platforms.

Education

University of Toronto | Ontario, Canada

Information Science Major, Bachelor of Information, Faculty of Information

09/2025 – 07/2027

Athabasca University | Alberta, Canada

Computer Science

A+

Broward College | FL, USA

Associate of Applied Science Degree in Software Development

GPA: 3.94

Personal Projects

Liver Tumor Analysis

Duration: 07/2023 – 08/ 2023

Repository: <https://github.com/Pintopie/Liver-Tumor-ML>

- Designed and implemented an end-to-end machine-learning pipeline to detect liver tumours from MRI scans using a public Kaggle dataset; developed reproducible Jupyter notebooks and preprocessing scripts to standardize inputs across sequences.
- Performed comprehensive image preprocessing and augmentation (intensity normalization, resizing, histogram equalization, noise reduction) and engineered domain-aware features (texture, histogram, morphological descriptors) for classical ML models.

- Trained and compared multiple classifiers (e.g., Random Forest, SVM, Gradient Boosting) using stratified cross-validation and hyperparameter tuning (GridSearchCV); evaluated models on precision/recall, F1-score and ROC-AUC to select the best production candidate.
- Implemented robust data pipelines and experiment tracking to ensure reproducibility; documented data cleaning decisions, feature transforms, and model selection criteria in the repo.
- Technologies: Python, NumPy, Pandas, scikit-image / OpenCV, scikit-learn, Matplotlib/Seaborn, Jupyter, Docker (optional), Kaggle dataset.

Local AI Model with RAG Pipeline

Duration: 08/2024 – 03/2025

Repository: <https://github.com/Pintopie/local-rag-system>

- Engineered a local Retrieval-Augmented Generation (RAG) pipeline with a React-based UI to enable private, low-latency conversational search over local documents and knowledge bases.
- Built a lightweight vector-retrieval stack (document chunking, embedding generation, and nearest-neighbor search) and integrated an on-device inference engine (Ollama) to serve responses without external API calls-preserving data privacy and reducing latency.
- Implemented conversational context management (query rewriting, context windowing, prompt templates) and caching to improve relevance and responsiveness; implemented frontend features for threaded conversations and source attribution.
- Focused on efficient local storage and retrieval (compact embeddings, batching, incremental indexing) and end-to-end UX-enabling RAG workflows for offline or sensitive-data scenarios.
- Technologies: React (TypeScript), Ollama (local LLM), embeddings (sentence-transformers or local models), FAISS, Python FastAPI, Docker.

Enterprise Projects

HealthBridge AI – Back End Engineer

JarvisMD AI Clinical Assistant

Duration: 11/2024 – 04/2025

- Backend engineering for an enterprise-grade AI medical transcription and assistant platform (JarvisMD) used to ingest clinician audio, run automated transcription pipelines, and deliver LLM-powered clinical summaries and structured notes.
- Implemented a scalable, containerized microservice architecture using Docker and docker-compose; designed production runtime with Gunicorn + Uvicorn workers and Nginx reverse-proxy for high availability and traffic routing (compose-managed multi-replica backend).
- Built robust startup and migration automation: prestart DB readiness checks with tenacity, automated Alembic migrations, and deterministic initial data seeding to enable reliable deployments and CI/CD-friendly rollouts.
- Integrated enterprise SSO (Keycloak) and JWT-based auth flows, enforced role-based access, and implemented secure secret management via environment files to meet internal compliance and data-privacy constraints.
- Developed audio ingestion and processing pipeline (FFmpeg, pydub, Amazon Transcribe + a dedicated transcribe service), persisted recordings and AI artifacts in Azure Blob Storage and MongoDB, and optimized storage/IO for concurrent sessions.
- Integrated LLM workflows using OpenAI and LangChain libraries to provide automated summarization, clinical reasoning and structured outputs; persisted AI metadata and session artifacts in MongoDB for auditability.
- Improved observability and reliability: structured logging, log rotation, and mounting persistent log volumes; added health checks and retry strategies to reduce downtime in orchestrated environments.
- Collaborated with cross-functional teams (frontend, prompt-manager, DevOps, and PO) to align on secure deployment, regulatory compliance, and performance targets; prepared runbooks and developer docs to accelerate onboarding and reduce time-to-first-deploy.

Technologies: Python 3.11, FastAPI, Uvicorn/Gunicorn, NodeJs for microservice, SQLAlchemy & Alembic, PostgreSQL & MongoDB, KeyCloak (OIDC), Docker, Nginx, FFmpeg, Amazon Transcribe, Azure Blob Storage, OpenAI (LangChain), pydub, pytest (tests), logging/tenacity for reliability.

Netpalm – Software Developer Intern

Production Outage Monitor (Company Internal)

Duration: 2 weeks

- Designed and implemented an internal production outage monitoring & notification service using Python and the Zulip API to deliver real-time alerts to engineering teams.
- Built health-check and heartbeat probes for production servers and services; implemented multi-layer detection (HTTP checks, process & port checks, and synthetic transactions) to reduce false positives and rapidly triage incidents.
- Implemented an alerting pipeline that aggregates status, deduplicates events, applies severity policies, and sends actionable Zulip messages (with service, host, timestamp, and remediation hints) to designated Zulip streams and team channels.
- Deployed the service to AWS (EC2) with secure environment configuration.
- Added retry/backoff logic, exponential backoff and circuit-breaker patterns to handle transient network issues; persisted recent event history for deduplication and incident correlation.
- Improved observability: integrated structured logging, metrics (Prometheus) and basic dashboards/alerts to track uptime trends and MTTR.
- Collaborated with DevOps and SRE to define alert thresholds, on-call routing, and runbooks for common failure modes; reduced mean time to acknowledgement (MTTA) through targeted automations and clear alert payloads.

Technologies: Python, Zulip API, AWS (EC2, Secrets Manager / SSM, CloudWatch), HTTP health checks, Prometheus, Docker, basic persistence (Redis cache) for deduplication, Git.

Netpalm – Software Developer Intern

Production Automation Test

Duration: 10/2024 – 12/2024

- Built and maintained end-to-end automation test suites using Selenium to validate front-end functionality and UI flows across browsers and screen sizes.
- Created and executed comprehensive API test collections in Postman to validate backend endpoints, enforce contract stability, and automate regression checks.
- Integrated Selenium and Postman tests into CI/CD pipelines to run on each pull request and nightly builds, enabling early detection of regressions and faster release cycles.
- Collaborated with QA and developers to triage failures, prioritize fixes, and iterate on flaky tests, improving test reliability and reducing false positives.

Technologies: Selenium, Postman, CI/CD (GitHub Actions / Jenkins).

Certifications

- Victorian Certificate of Education
- IELTS 7.5/9.0

External Links

- <https://www.linkedin.com/in/kenny-nguyen-42ba2624b>
- <https://github.com/Pintopic>